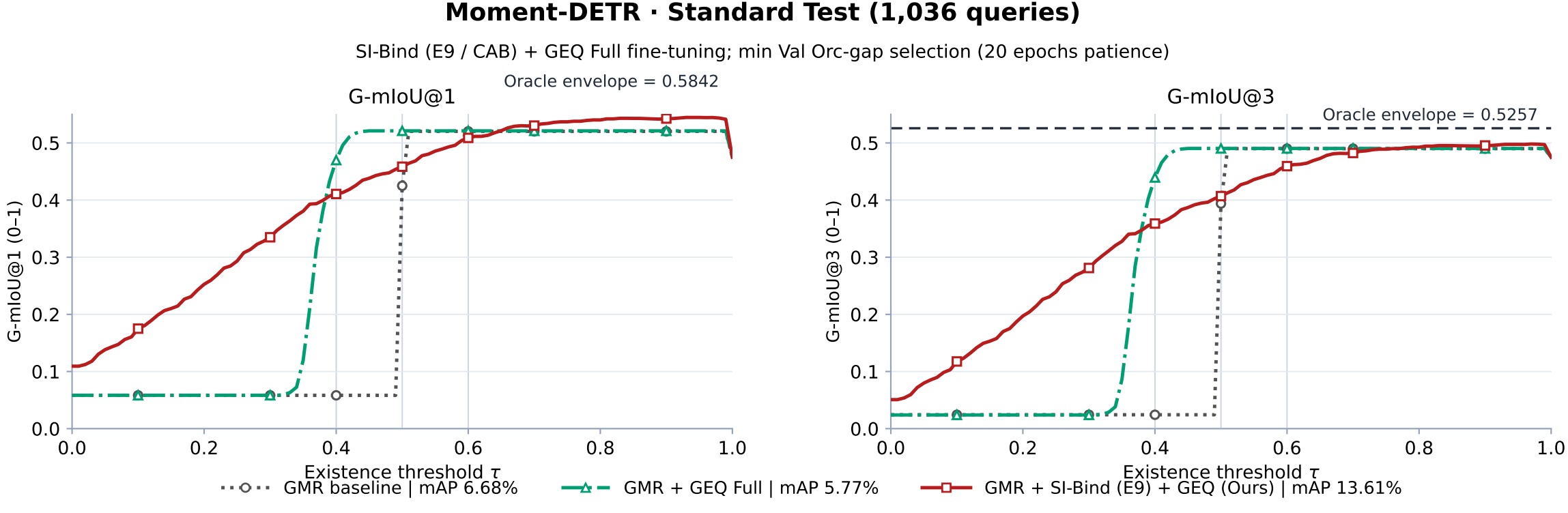
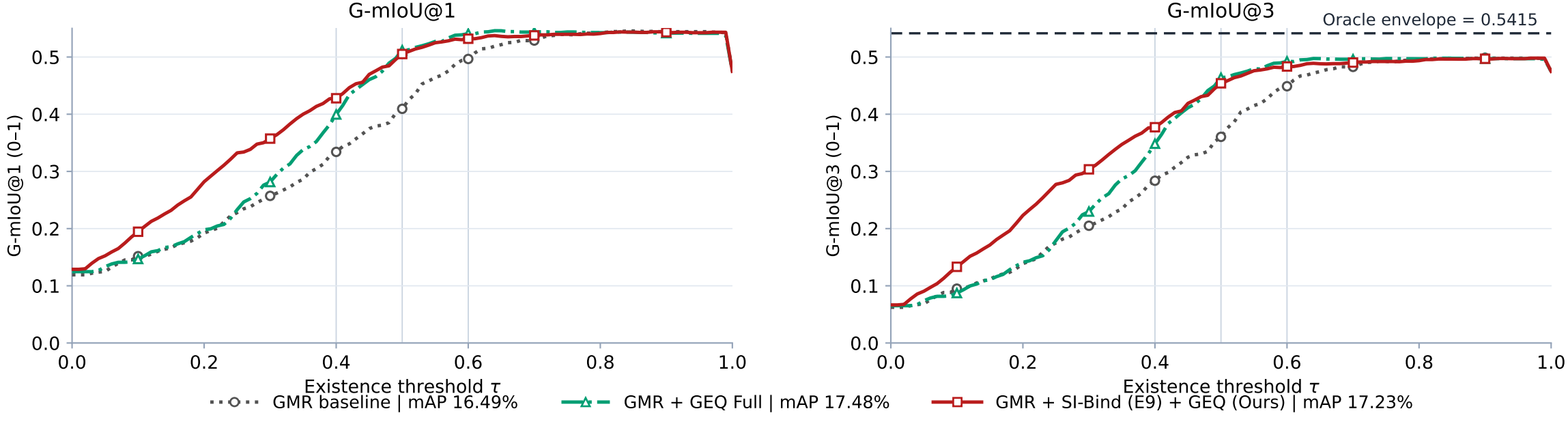

Moment-DETR · Standard Test (1,036 queries)
SI-Bind (E9 / CAB) + GEQ Full fine-tuning; min Val Orc-gap selection (20 epochs patience)
Oracle envelope = 0.5842
G-mIoU@1
G-mIoU@1 (0–1)
Existence threshold τ
Oracle envelope = 0.5257
G-mIoU@3
G-mIoU@3 (0–1)
Existence threshold τ
GMR baseline | mAP 6.68%
GMR + GEQ Full | mAP 5.77%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 13.61%
ΔmAP vs GMR: +6.93 pp (vs GEQ: +7.84 pp)
ΔG1 mean vs GMR (.4/.5/.6): +12.48 pp
ΔG3 mean vs GMR (.4/.5/.6): +10.57 pp

EaTR · Standard Test (1,036 queries)

SI-Bind (E9 / CAB) + GEQ Full fine-tuning; min Val Orc-gap selection (20 epochs patience)

Oracle envelope = 0.6037



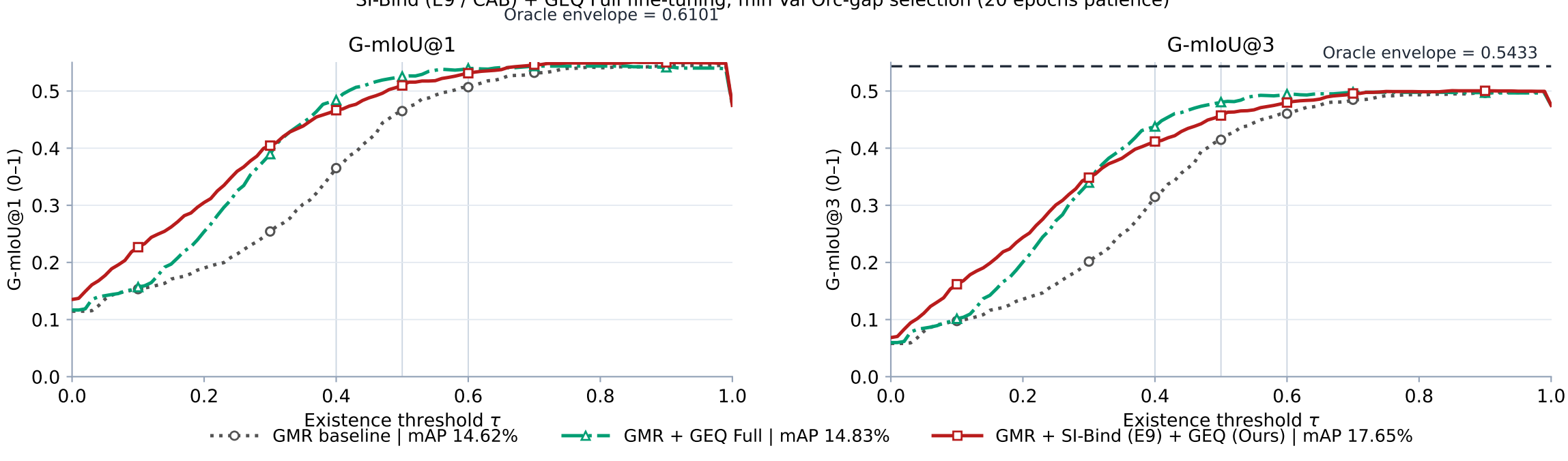
ΔmAP vs GMR: +0.74 pp (vs GEQ: -0.25 pp)

ΔG1 mean vs GMR (.4/.5/.6): +7.50 pp

ΔG3 mean vs GMR (.4/.5/.6): +7.39 pp

QD-DETR · Standard Test (1,036 queries)

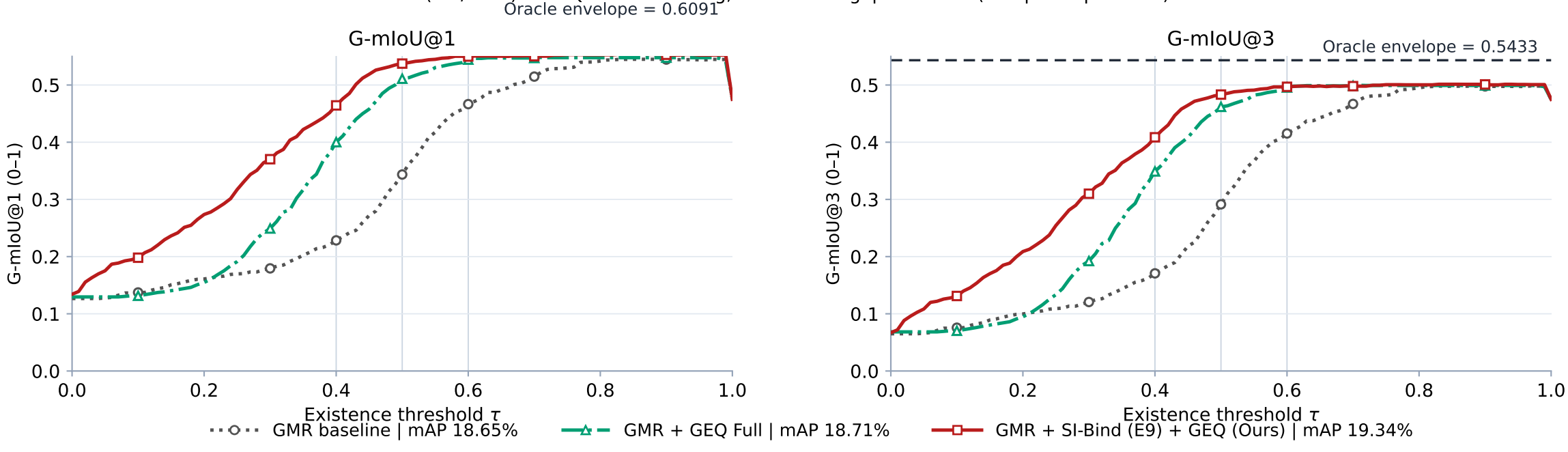
SI-Bind (E9 / CAB) + GEQ Full fine-tuning; min Val Orc-gap selection (20 epochs patience)



ΔmAP vs GMR: +3.03 pp (vs GEQ: +2.82 pp) **ΔG1 mean vs GMR (.4/.5/.6): +5.69 pp** **ΔG3 mean vs GMR (.4/.5/.6): +5.30 pp**

CG-DETR · Standard Test (1,036 queries)

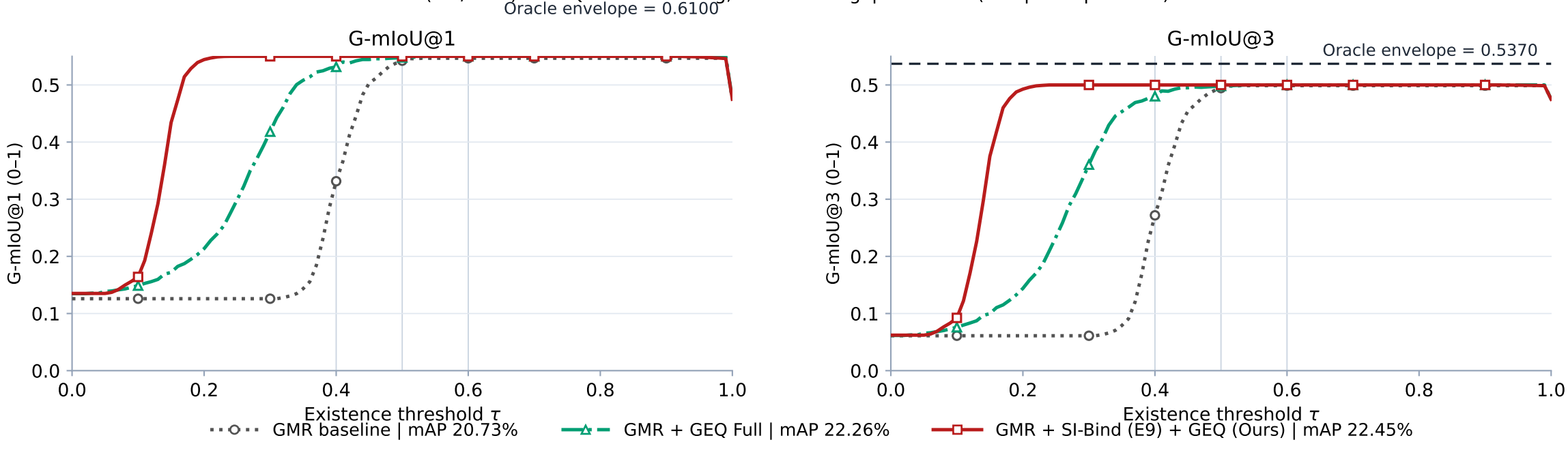
SI-Bind (E9 / CAB) + GEQ Full fine-tuning; min Val Orc-gap selection (20 epochs patience)



ΔmAP vs GMR: +0.69 pp (vs GEQ: +0.63 pp) **ΔG1 mean vs GMR (.4/.5/.6): +17.10 pp** **ΔG3 mean vs GMR (.4/.5/.6): +17.04 pp**

Sim-DETR · Standard Test (1,036 queries)

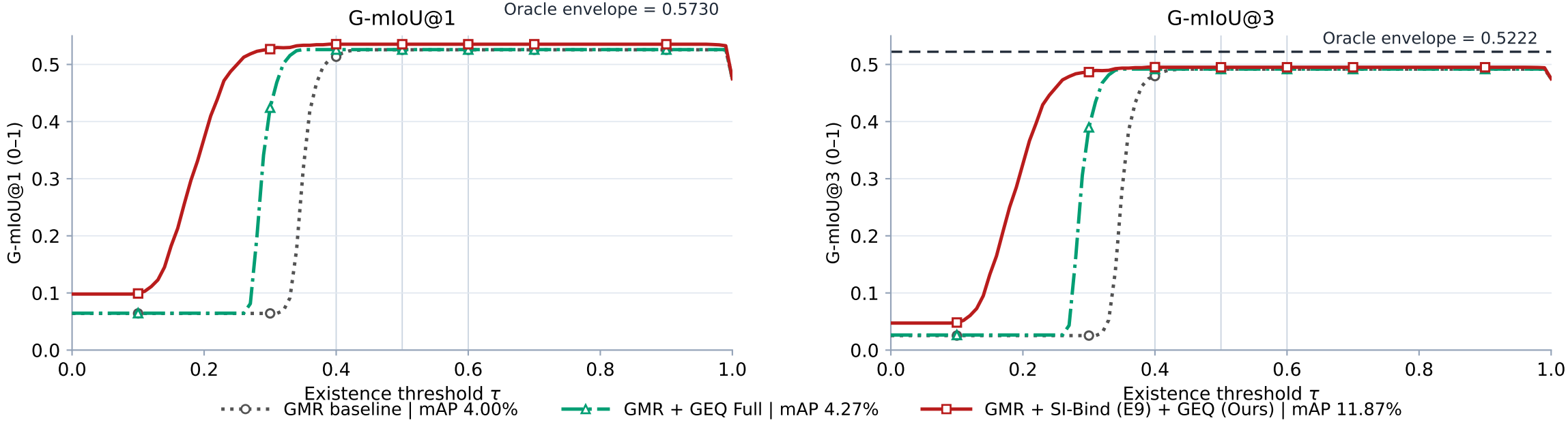
SI-Bind (E9 / CAB) + GEQ Full fine-tuning; min Val Orc-gap selection (20 epochs patience)



ΔmAP vs GMR: +1.72 pp (vs GEQ: +0.19 pp) **ΔG1 mean vs GMR (.4/.5/.6): +7.65 pp** **ΔG3 mean vs GMR (.4/.5/.6): +7.83 pp**

BM-DETR · Standard Test (1,036 queries)

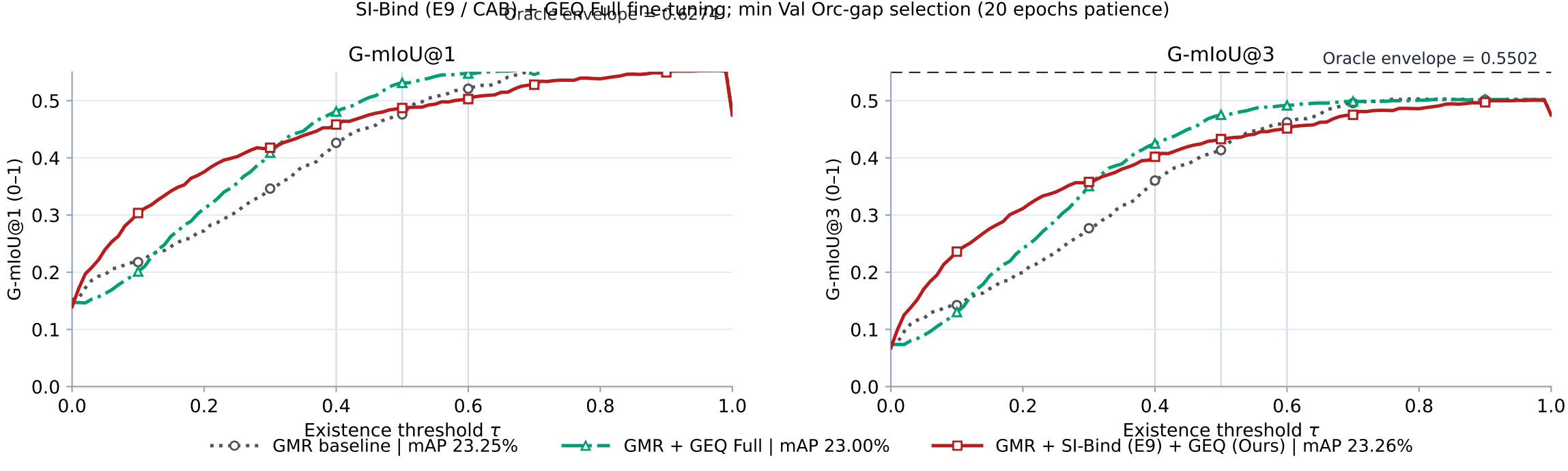
SI-Bind (E9 / CAB) + GEQ Full fine-tuning; min Val Orc-gap selection (20 epochs patience)



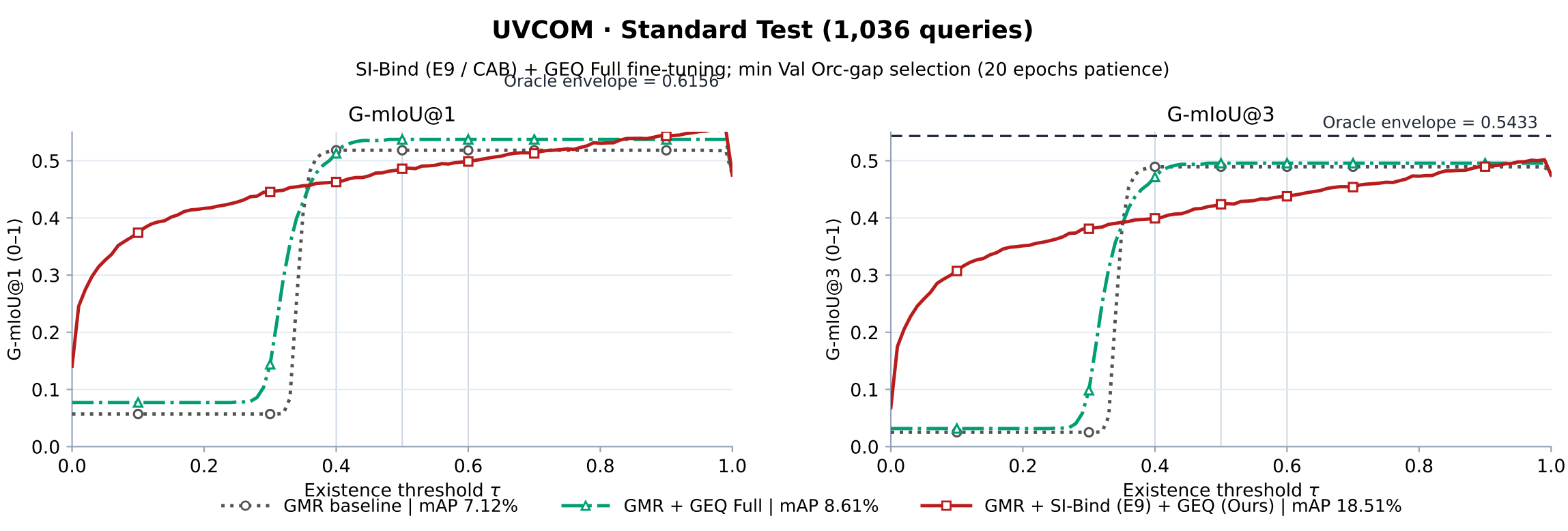
ΔmAP vs GMR: +7.87 pp (vs GEQ: +7.60 pp) **ΔG1 mean vs GMR (.4/.5/.6): +1.38 pp** **ΔG3 mean vs GMR (.4/.5/.6): +0.74 pp**

BAM-DETR · Standard Test (1,036 queries)

SI-Bind (E9 / CAB) + GEQ Full fine-tuning; min Val Orc-gap selection (20 epochs patience)



ΔmAP vs GMR: +0.01 pp (vs GEQ: +0.26 pp) **ΔG1 mean vs GMR (.4/.5/.6): +0.84 pp** **ΔG3 mean vs GMR (.4/.5/.6): +1.67 pp**


UVCOM · Standard Test (1,036 queries)
SI-Bind (E9 / CAB) + GEQ Full fine-tuning; min Val Orc-gap selection (20 epochs patience)
Oracle envelope = 0.6156
G-mIoU@1
G-mIoU@1 (0–1)
Existence threshold τ
Oracle envelope = 0.5433
G-mIoU@3
G-mIoU@3 (0–1)
Existence threshold τ
GMR baseline | mAP 7.12%
GMR + GEQ Full | mAP 8.61%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 18.51%
ΔmAP vs GMR: +11.39 pp (vs GEQ: +9.90 pp)
ΔG1 mean vs GMR (.4/.5/.6): -3.58 pp
ΔG3 mean vs GMR (.4/.5/.6): -6.91 pp